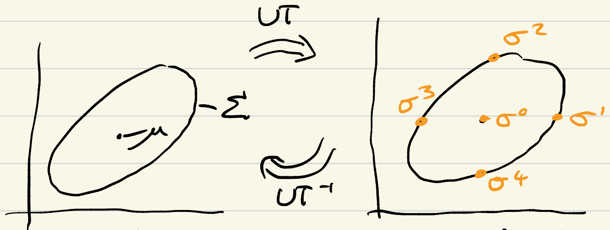
Recall Unscented Transform (or Sigma Transform)



(μ, Σ)

$\sigma^i = (x^i, w^i)$

$\{\sigma^i\}_{i=0}^{2n}$

Recall UT and UT^{-1}

Forward Transform:

free parameter

$$UT(\mu, \Sigma) = \begin{cases} x^0 = \mu, \quad w^0 = \frac{\lambda}{\lambda+n} \\ x^i = \mu + \left(\sqrt{(\lambda+n)\Sigma}\right)_i \\ \quad w^i = \frac{1}{2(\lambda+n)}, \; i = 1, \dots, n \\ x^{n+i} = \mu - \left(\sqrt{(\lambda+n)\Sigma}\right)_i \\ \quad w^{n+i} = \frac{1}{2(\lambda+n)}, \; i = 1, \dots, n \end{cases}$$

col of mat.

where $\left(\sqrt{(\lambda+n)\Sigma}\right)_i$ is the ith column of matrix square root $\sqrt{(\lambda+n)\Sigma}$

Inverse Transform:

$$UT^{-1}\left(\{x^i, w^i\}_{i=0}^{2n}\right) = \begin{cases} \mu = \sum_{i=0}^{2n} w^i x^i \\ \Sigma = \sum_{i=0}^{2n} w^i (x^i - \mu)(x^i - \mu)^T \end{cases}$$

Verify that $UT^{-1}(UT(\mu, \Sigma)) = (\mu, \Sigma)$

(we actually get back (μ, Σ) from σ-points)

mean:

$$\mu \overset{?}{=} \sum_{i=0}^{2n} w^i x^i = \frac{\lambda}{\lambda+n}\mu + \sum_{i=1}^{n} \frac{1}{2(\lambda+n)}\left(\mu + \cancel{\left(\sqrt{(\lambda+n)\Sigma}\right)_i}\right) + \sum_{i=1}^{n} \frac{1}{2(\lambda+n)}\left(\mu - \cancel{\left(\sqrt{(\lambda+n)\Sigma}\right)_i}\right)$$

$$= \frac{\lambda}{\lambda+n}\mu + \frac{n}{2(\lambda+n)}\mu + \frac{n}{2(\lambda+n)}\mu$$

$$= \left(\frac{\lambda}{\lambda+n} + \frac{n}{\lambda+n}\right)\mu \overset{\checkmark}{=} \mu \quad \square$$

covariance:

$$\Sigma \overset{?}{=} \sum_{i=0}^{2n} w^i (x^i - \mu)(x^i - \mu)^T$$

$$= \frac{\lambda}{\lambda+n}\overset{0}{\cancel{(\mu - \mu)(\mu - \mu)^T}} + \sum_{i=1}^{n} \frac{1}{2(\lambda+n)} \Big(\overbrace{\cancel{\mu} + \left(\sqrt{(\lambda+n)\Sigma}\right)_i}^{x^i} - \cancel{\mu}\Big) \times (\dots)^T + \sum_{i=1}^{n} \frac{1}{2(\lambda+n)} \Big(\underbrace{\cancel{\mu} - \left(\sqrt{(\lambda+n)\Sigma}\right)_i}_{x^{n+i}} - \cancel{\mu}\Big)(\dots)^T$$

$$= \sum_{i=1}^{n} \frac{1}{\cancel{(\lambda+n)}} \cancel{\sqrt{\lambda+n}}\left(\sqrt{\Sigma}\right)_i \times \cancel{\sqrt{\lambda+n}}\left(\sqrt{\Sigma}\right)_i^T$$

$$= \sum_{i=1}^{n} \left(\sqrt{\Sigma}\right)_i \left(\sqrt{\Sigma}\right)_i^T \overset{?}{=} \Sigma$$

recall for matrix $M = \begin{bmatrix} | & & | \\ m_1 & \cdots & m_n \\ | & & | \end{bmatrix}$

we have:

$$MM^T = \begin{bmatrix} | & & | \\ m_1 & \cdots & m_n \\ | & & | \end{bmatrix} \begin{bmatrix} - & m_1^T & - \\ & \vdots & \\ - & m_n^T & - \end{bmatrix}$$

$$= \sum_{i=1}^{n} m_i m_i^T$$

for us, we have

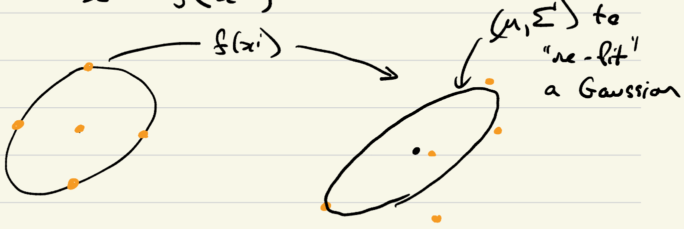
$$\sum_{i=1}^{n} \left(\sqrt{\Sigma}\right)_i \left(\sqrt{\Sigma}\right)_i^T = \left(\sqrt{\Sigma}\right)\left(\sqrt{\Sigma}\right)^T = \Sigma$$

by def. of mat. sqrt $\sqrt{\Sigma}$

$\square$

σ-points give a point-wise "linearization":

$$\hat{x}^i = f(x^i)$$



UKF Algo

init: $X_0 \sim \mathcal{N}(\mu_{0|0}, \Sigma_{0|0})$

predict:

$$UT(\mu_{t-1|t-1}, \Sigma_{t-1|t-1}) \rightarrow \{x^i_{t-1|t-1}, w^i\}_{i=0}^{2n}$$

predict each σ-point: $$\boxed{\bar{x}^i_{t|t-1} = f(x^i_{t-1|t-1}, u_{t-1})}$$ (no noise yet)

(w^i do not change)

$UT^{-1}(\{\bar{x}^i_{t|t-1}, w^i\}_{i=0}^{2n})$:

$$\boxed{\mu_{t|t-1} = \sum_{i=0}^{2n} w^i \bar{x}^i_{t|t-1}}$$

$$\boxed{\Sigma_{t|t-1} = \sum_{i=0}^{2n} w^i (\bar{x}^i_{t|t-1} - \mu_{t|t-1})(\ldots)^T + Q_{t-1}}$$

add noise cov. here.

update:

$$UT(\mu_{t|t-1}, \Sigma_{t|t-1}) \rightarrow \{x^i_{t|t-1}, w^i\}_{i=0}^{2n}$$

predict σ-point measurement: $$\boxed{y^i_{t|t-1} = g(x^i_{t|t-1})}$$

σ-point expected measurement: $$\boxed{\hat{y}_{t|t-1} = \sum_{i=0}^{2n} w^i y^i_{t|t-1}}$$

σ-point measurement covariance: $$\boxed{\Sigma_y = \sum_{i=0}^{2n} w^i (y^i_{t|t-1} - \hat{y}_{t|t-1})(\ldots)^T + R_t}$$

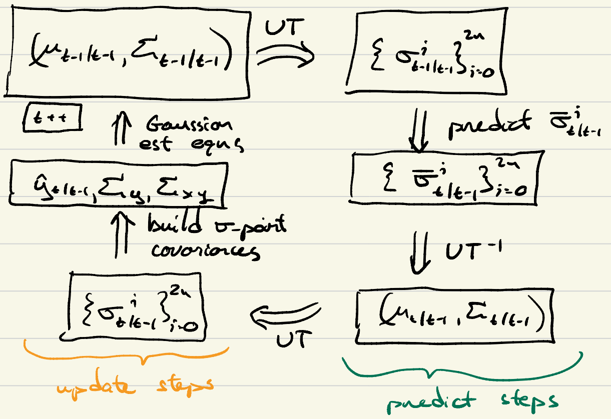
add meas. noise cov.

σ-point cross covariance: $$\boxed{\Sigma_{xy} = \sum_{i=0}^{2n} w^i (x^i_{t|t-1} - \mu_{t|t-1}) \times (y^i_{t|t-1} - \hat{y}_{t|t-1})^T}$$

compute Gaussian est. eqns: $$\boxed{\begin{aligned} \mu_{t|t} &= \mu_{t|t-1} + \Sigma_{xy}\Sigma_y^{-1}(y_t - \hat{y}_{t|t-1}) \\ \Sigma_{t|t} &= \Sigma_{t|t-1} - \Sigma_{xy}\Sigma_y^{-1}\Sigma_{yx} \end{aligned}}$$

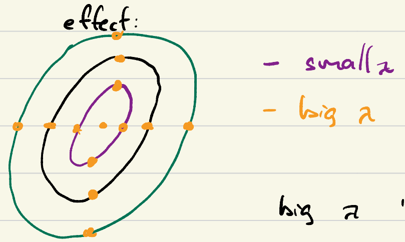
return: $\boxed{(\mu_{t|t}, \Sigma_{t|t})}$

Notice: No Jacobians required. Only point evaluations of $f(\cdot), g(\cdot)$ functions.


$(\mu_{t-1|t-1}, \Sigma_{t-1|t-1})$
UT
$\{\sigma^i_{t-1|t-1}\}_{i=0}^{2n}$
predict $\bar{\sigma}^i_{t|t-1}$
$\{\bar{\sigma}^i_{t|t-1}\}_{i=0}^{2n}$
UT^{-1}
$(\mu_{t|t-1}, \Sigma_{t|t-1})$
UT
$\{\sigma^i_{t|t-1}\}_{i=0}^{2n}$
build σ-point covariances
$\hat{y}_{t|t-1}, \Sigma_y, \Sigma_{xy}$
Gaussian est eqns
t++
update steps
predict steps

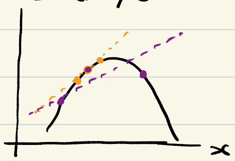
1. λ parameter: usually people chose λ = 2

effect:



big λ "smooths out" nonlinearity

1-D example



2. Matrix Square Roots:

def.: $\sqrt{\Sigma}\sqrt{\Sigma}^T = \Sigma$

many choices:

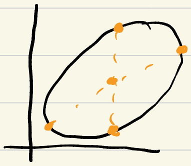
(a) Cholesky Sqrt:

$$\Sigma = \underset{\uparrow \text{ lower triangular}}{L} L^T$$

$$\sqrt{\Sigma} = L = \begin{bmatrix} | & & 0 \\ | & | & \\ | & | & \ddots \end{bmatrix}$$

1 col, 2 col, n^{th} col

$(\sqrt{\Sigma})_i$



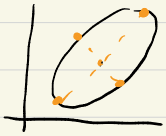
messy but fast to compute.

(b) Ellipse-Aligned

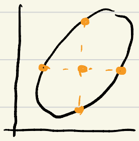
$$\Sigma = \underset{\text{eig. vects}}{U}\,\underset{\text{diag. eigs.}}{\Lambda}\,U^T$$

define: $\sqrt{\Sigma} = U\Lambda^{1/2}$

$$\sqrt{\Sigma}\sqrt{\Sigma}^T = U\overset{\Lambda}{\cancel{\Lambda^{1/2}\Lambda^{1/2}}}U^T = \Sigma$$



(c) Axis-Aligned



$$\Sigma = U\Lambda U^T$$

def: $\sqrt{\Sigma} = U\Lambda^{1/2}U^T$

$$\sqrt{\Sigma}\sqrt{\Sigma}^T = (U\Lambda^{1/2}\overset{1}{\cancel{U^T})(U}\Lambda^{1/2}U^T) = U\Lambda U^T = \Sigma$$

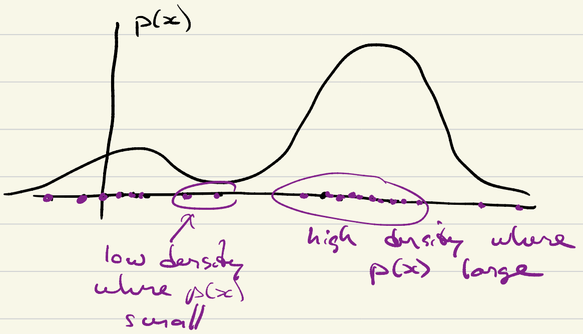
Particle Filter (PF)

Main Idea: represent distribution as a set of weighted samples (a.k.a. particles)

$$\{(x^i, w^i)\}_{i=1}^N$$

N usually ~ 10-100 per dimension

so $\mathbb{R}^N \Rightarrow 10^N - 100^N$ particles



motivated by Law of Large Numbers:

$$E[f(X)] \approx \frac{1}{N}\sum_{i=1}^N f(X^i), \quad X^i \sim p(x) \text{ i.i.d.}$$

here $w^i = \frac{1}{N}$

Q: what if we have

$$X^i \sim q(x) \quad \text{i.i.d.}$$

↑ proposal distribution

call $p(x)$ target distribution.

Can we still find approximations

$E_p[f(X)]$ as a function of $X^i \sim q(x)$ i.i.d.

Called Importance Sampling.

Consider:

$$E_p[f(X)] = \int_x f(x)\, p(x)\, dx$$

$$= \int_x f(x) \underbrace{\frac{p(x)}{q(x)}}_{w(x) - \text{weighting function}} q(x)\, dx$$

$$= \int_x f(x)\, w(x)\, q(x)\, dx$$

$$= E_q[f(X)\, w(X)]$$

$$\approx \sum_{i=1}^N w^i f(X^i), \quad X^i \sim q(x) \text{ i.i.d.}$$

and $w^i = w(X^i)$

Claim: for Bayes' rule, we can get w_i without knowing $q(x)$ or $p(x)$

let $q(x)$ be prior

and $p(x|y)$ be posterior

$$p(x|y) = \frac{p(y|x)\, q(x)}{p(y)}$$

recall weighting function

$$w(x) = \frac{p(x|y)}{q(x)} = \frac{p(y|x)}{p(y)}$$

$$w(x) \propto p(y|x)$$

↑ proportional to

so given $Y = y$, $X = x^i$

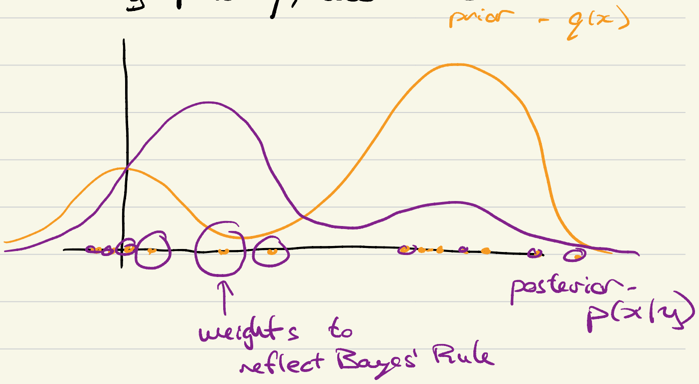
$X \sim q(x)$ prior

unnormalized → $\boxed{\hat{w}^i = p(Y = y \mid X = x^i)}$

normalized → $\boxed{w^i = \frac{\hat{w}^i}{\sum_{j=1}^N \hat{w}^i}}$

and posterior $p(x|y)$ is represented by $\boxed{\{x^i, w^i\}_{i=1}^N}$

graphically, we have



So we can approximate, e.g., posterior mean as

$$\mu_{x|y} = E[X \mid Y] \approx \sum_{i=1}^N w^i x^i$$

where $x^i \sim q(x)$

and w^i as above

Particle Filter Algorithm

(Sequential Importance Resampling, SIR Filter)

Init: particle set, uniform weights

$$\{ x^i_{0|0}, w^i_0 = \tfrac{1}{N} \}_{i=1}^N , \quad x^i_{0|0} \sim p(x_0) \text{ i.i.d.}$$

predict:

$$\boxed{x^i_{t|t-1} = f(x^i_{t-1|t-1}, u_{t-1}) + W^i_{t-1}}$$

$$\boxed{\text{where } W^i_{t-1} \sim p_W(w_{t-1})}$$

process noise dist.

predict each particle with different random noise draw

(no change to weights)

update:

$$\boxed{\hat{w}^i_t = p(Y_t = y_t \mid X_t = x^i_{t|t-1})}$$

find measurement likelihood for each particle $x^i_{t|t-1}$

$$\boxed{w^i_t = \frac{\hat{w}^i}{\sum_{i=1}^{N} \hat{w}^i}}$$

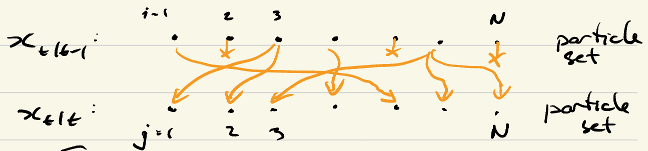
normalize

importance resample:

$$\{ x^j_{t|t}, \underbrace{w^j_t = \tfrac{1}{N}}_{\text{uniform weights}} \}_{j=1}^N \Leftarrow \{ x^i_{t|t-1}, w^i_t \}_{i=1}^N$$

Importance Resampling (IR)

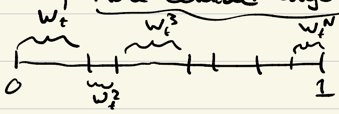
main idea: choose a new set of particles, $x^j_{t|t}$ by selecting $x^i_{t|t-1}$ with prob. w^i_t.



```
for $j=1$ to $N$
    $x^j_{t|t}$ draw from $\{x^i_{t|t-1}\}_{i=1}^N$
        where $x^j_{t|t} = x^i_{t|t-1}$ with prob $w^i_t$
end
```

More detailed algorithm for IR

1. bins of width w^i



2. draw random scalar s_j uniformly on $[0,1]$
3. if s_j lands in bin i, assign $x^j_{t|t} = x^i_{t|t-1}$
4. repeat for $j = 1, \ldots, N$.